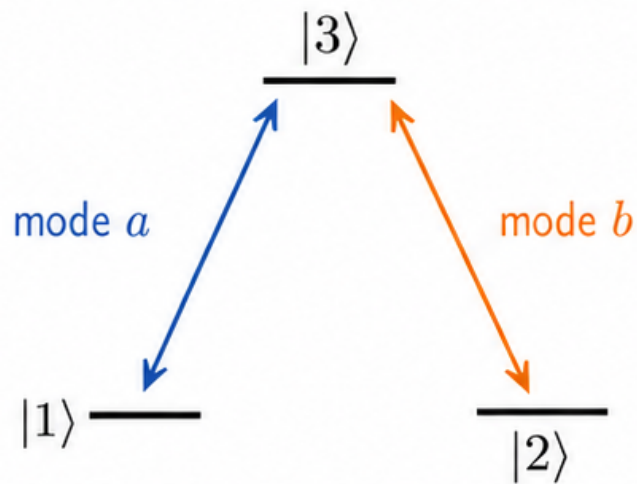
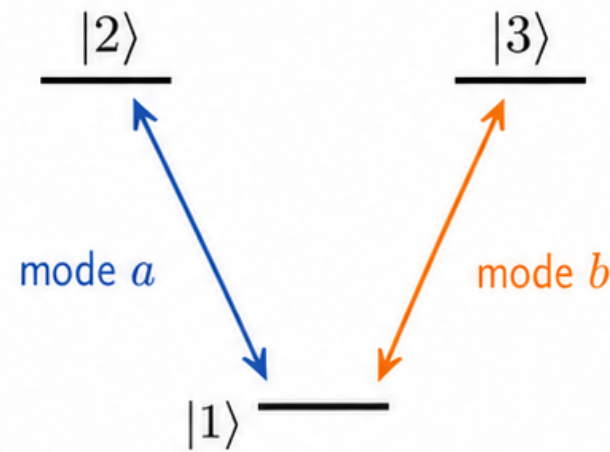
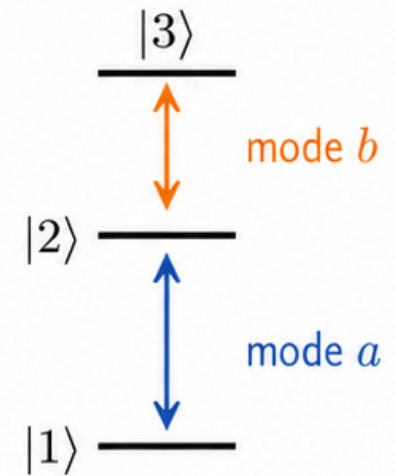


(a) Λ configuration

(b) V configuration

(c) ladder Ξ configuration

(d)

same-sign detunings

 $(+,+)$ or $(-,-)$

$$\Delta_a - \Delta_b \approx 0$$

opposite-sign detunings

 $(+,-)$ or $(-,+)$

$$\Delta_a + \Delta_b \approx 0$$

 Λ

$$a^\dagger b \sigma_{12} + \text{H.c.}$$

lower-manifold exchange / mode conversion

0

V

$$-a^\dagger b \sigma_{32} + \text{H.c.}$$

upper-manifold exchange / mode conversion

0

 Ξ

0

$$a^\dagger b^\dagger \sigma_{13} + \text{H.c.}$$

pair creation / two-photon absorption

$$b = a \Rightarrow a^{\dagger 2} \sigma_{13} + \text{H.c.}$$

after atomic reduction: $a^{\dagger 2} + a^2$ self-commutators $\rightarrow$ photon-number-dependent AC Stark shifts $\rightarrow$ corrected resonance